SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – SIH1733

Problem Statement Title – SAR Image Colorization for Comprehensive Insight using Deep Learning Model

Theme – Space Technology

PS Category – Software

Team Number – 71

Team Name – Delta Force

Delta Force

IDEA TITLE

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ChromaSAR — colorized SAR you can act on, with a confidence you can check

- Problem: optical satellites go blind under cloud — exactly during the monsoon. Radar sees through it, but radar is grayscale texture only specialists can read.
- Solution: a conditional GAN colorizes Sentinel-1 SAR into optical-like imagery, trained on 10,000 co-registered SAR–optical pairs.
- USP — radar carries no colour, so every pixel ships a CALIBRATED CONFIDENCE. Below your threshold the output goes grey: “insufficient evidence”, not a confident guess. Per-pixel uncertainty on SAR is not new; what is new is that ours is calibrated, shown to the user, and GATES the downstream answer.
- That confidence GATES everything downstream — flood extent, change detection and surface cover are all filtered by it.
- Unique 1 — we diagnosed WHY existing models look unsatisfactory. L1 loss is minimised by the conditional MEAN of every plausible colour, which is blur by definition, and PSNR rewards the same thing so standard evaluation cannot see it. A gradient-difference loss fixed it: sharpness 0.21 → 0.78 of the real optical, saturation 45% → 102%.
- Unique 2 — we measure what does not work and REFUSE to ship it. Built-up land cover from radar alone scores AUC 0.483, below chance, so the app reports it as unavailable and says why.

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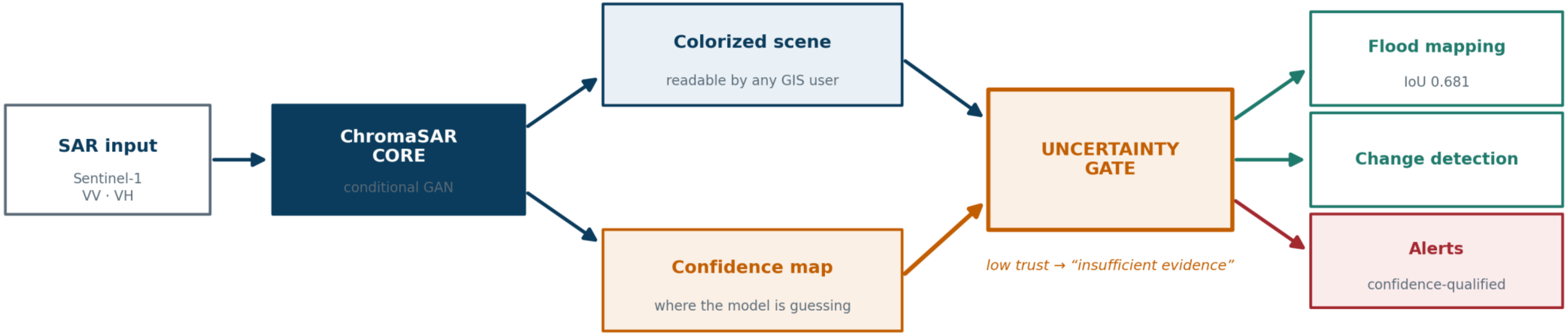
TECHNICAL APPROACH



Technologies

- Python · PyTorch · FastAPI · Next.js 15 · TypeScript · GDAL/rasterio · QGIS-ready GeoTIFF export
- Generator ResNet34-UNet (ImageNet encoder) + 70 × 70 PatchGAN discriminator
- Loss: L1 + frozen-VGG16 perceptual + adversarial + gradient-difference
- Confidence: 10-pass Monte-Carlo dropout, per pixel
- Flood: dual-polarisation threshold baseline + ResNet34-UNet segmentation, temperature-calibrated

Methodology

- Split by SCENE, not patch — adjacent tiles cannot leak across train/validation
- Trained on a serverless A10G GPU (~$3 total). Inference runs on CPU: 1.1 s per scene, in-process, no network call
- Prototype running today: 4 workspaces, upload your own file, georeferenced export, 43 regression tests green

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FEASIBILITY AND VIABILITY

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Feasibility — demonstrated, not projected

- Runs today on a laptop CPU. No GPU, no cloud dependency, no API key.
- Flood mapping IoU 0.681 on the Sen1Floods11 OFFICIAL test split, against 0.550 for the physics baseline. Probabilities calibrated: ECE 0.029 → 0.016.
- An uploaded file scores identically to our benchmark — 0.00% difference across three scenes on three continents.
- Honest positioning: 0.681 beats the Sen1Floods11 published baseline (0.662) and matches a supervised SAR-only model (0.676). SAR-only state of the art is ~0.72 (DeepSARFlood, IIT Delhi). We are competitive, not leading, and we say so.

Risks, and how we handle them

- Hallucination — the network invents colour the radar never carried. → Ship the confidence map and refuse below threshold. Surfaced, never hidden.
- Blur — L1 drives the output to the conditional mean. → Gradient-difference loss, plus a sharpness metric PSNR is structurally blind to.
- Flooded vegetation double-bounces and returns BRIGHT, not dark, so the physics threshold misses it. → Learned segmentation replaces physics where physics fails.
- Land cover from radar alone does not work (built-up AUC 0.483). → Measured, and refused. Cover is read from the co-registered Sentinel-2 optical and labelled as such.
- Domain shift across terrain. → Stratified sampling over agro-climatic zones, per-region fine-tuning.

IMPACT AND BENEFITS

Impact — target audience: Remote Sensing Image Analysts

- SAR interpretation drops from expert-only to any GIS user; exports open straight in QGIS on top of the source scene.
- Disaster response: flood extent mapped during the monsoon, exactly when optical satellites are blind. Permanent water is separated from new flooding — a river is not a disaster.

Benefits

- Social — faster, better-informed response in the months India floods most.
- Economic — removes dependence on cloud-free optical tasking; runs on existing hardware, no GPU purchase.
- Strategic — indigenous SAR-interpretation capability that transfers to RISAT and NISAR; works offline, in the field, with no connectivity.
- Environmental — the PS's own named uses: continuous monitoring of wetlands, glaciers and coastal change instead of seasonal snapshots.
- Archive unlock — decades of existing SAR holdings become visually searchable.

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RESEARCH AND REFERENCES

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Datasets

- Sen1Floods11 — Bonafilia et al., CVPRW 2020. 446 hand-labelled flood chips, 11 regions, official split. github.com/cloudtostreet/Sen1Floods11
- SEN1-2 — Schmitt et al., ISPRS 2018. 10,000 co-registered SAR–optical pairs; the paper names SAR colorization as an intended use. mediatum.ub.tum.de/1436631
- Imagery — Sentinel-1 / Sentinel-2 via ESA Copernicus. dataspace.copernicus.eu

Prior art and incumbents

- Copernicus GFM — operational global flood service. Already ships a per-pixel likelihood, but on a binary mask, with no colorization. extwiki.eodc.eu/gfm
- ISRO NRSC — near-real-time flood inundation via NDEM / Bhuvan (EOS-04, Sentinel-1). Authoritative, event-triggered, delivered as maps. nrsc.gov.in
- OSCAR, 2026 — uncertainty-aware SAR-to-optical diffusion. Closest prior art; the uncertainty stays in the training loss. arxiv.org/abs/2601.06835

Methods

- Isola et al., Image-to-Image Translation with Conditional Adversarial Networks (pix2pix), CVPR 2017 — our deliberate baseline.
- Mathieu et al., Deep Multi-Scale Video Prediction Beyond MSE, ICLR 2016 — the gradient-difference loss that removed our blur.
- Gal & Ghahramani, Dropout as a Bayesian Approximation, ICML 2016 — the basis of our per-pixel confidence.
- Guo et al., On Calibration of Modern Neural Networks, ICML 2017 — temperature scaling; ECE 0.029 → 0.016.
- Xu, Modification of NDWI (MNDWI), IJRS 2006 — optical water reference; we re-measured the threshold rather than adopting it.